



ANDERSON JUNIOR COLLEGE
HIGHER 2

2018 JC2 PRELIMINARY EXAMINATIONS

H2 BIOLOGY

9744/01

Paper 1 Multiple Choice

18 SEPT 2018
TUESDAY

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, PDG and identification number on the Answer Sheet.

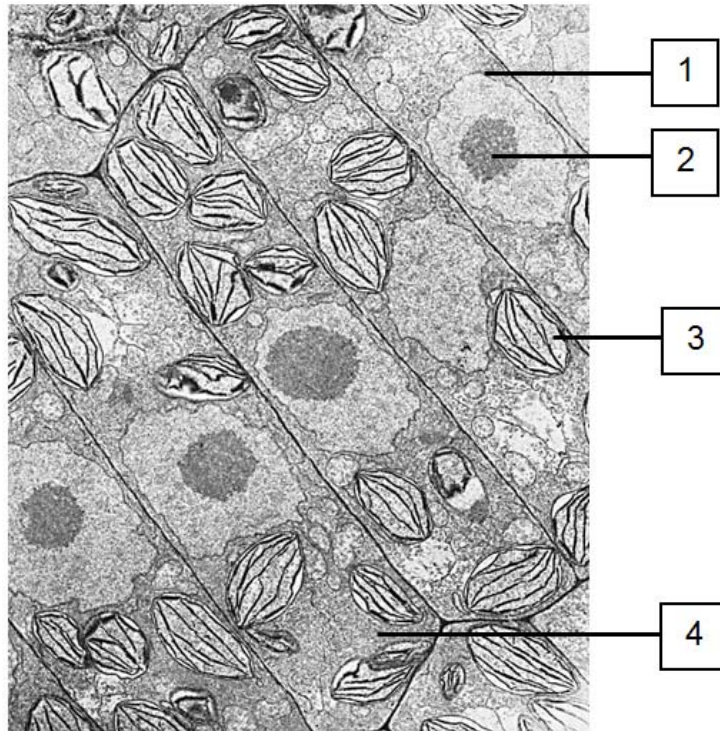
There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

The use of scientific calculators is expected, where appropriate.

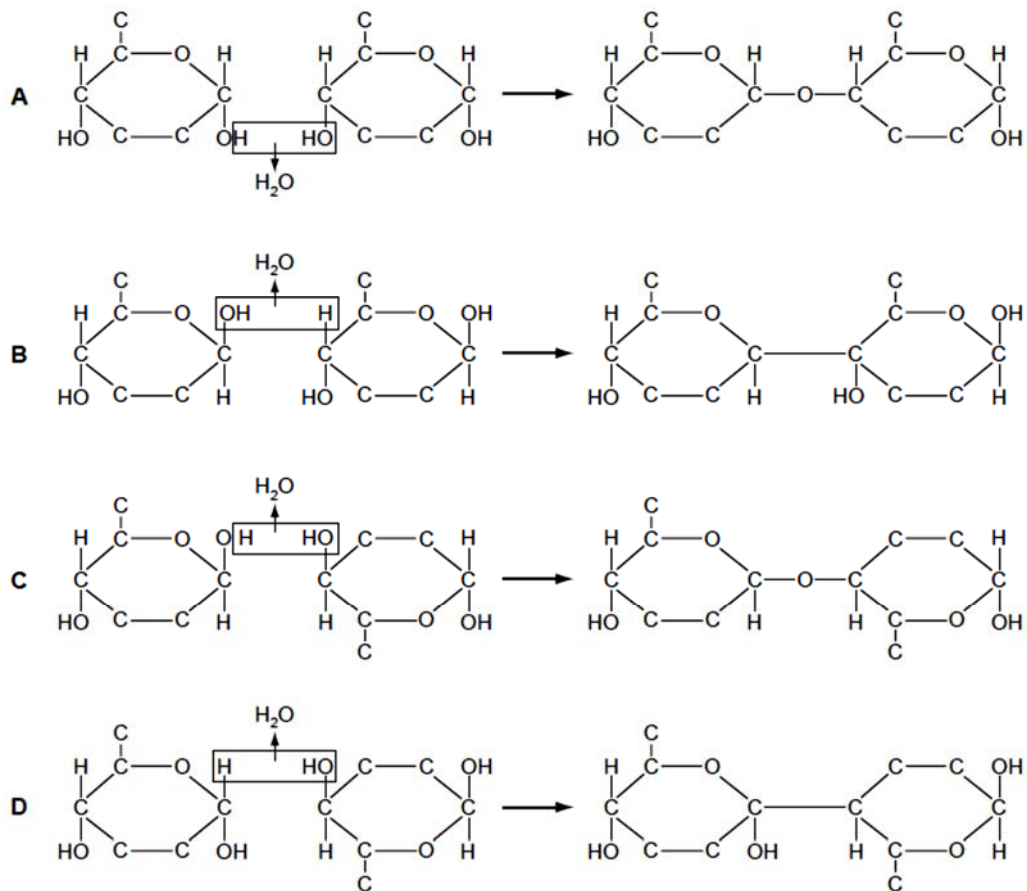
1 The electron micrograph shows cells from an aquatic plant.



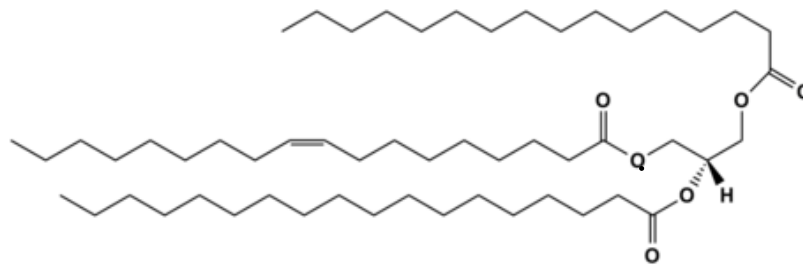
Which of the following about structures 1 to 4 is correct?

	contains DNA	contains rRNA	contains tRNA	contains proteins
A	1, 2	2	2, 3, 4	1, 4
B	1, 3, 4	2, 3	2, 4	3, 4
C	2, 3	2, 3, 4	3, 4	1, 2, 3, 4
D	2	3, 4	1, 3	1, 2

2 Which diagram shows the linking of two monomers together to form cellulose?



3 The diagram shows the structure of a lipid molecule.

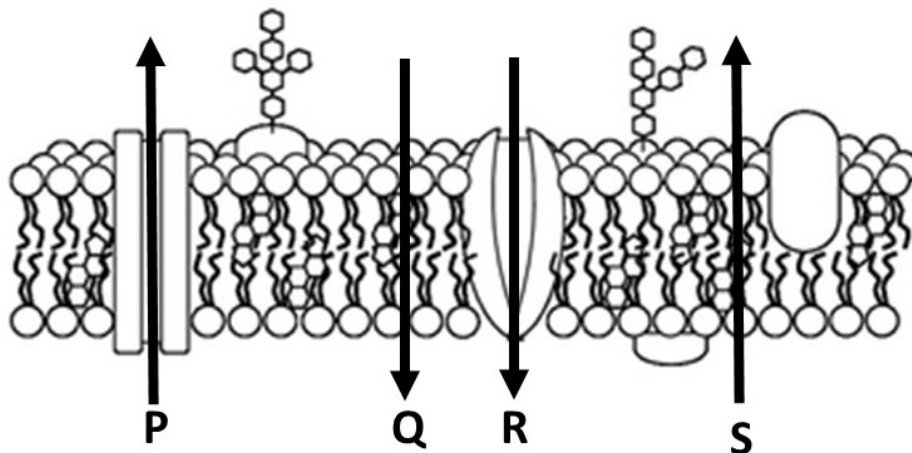


Which statements are **not** correct?

- 1 The fatty acid occupying the central (second) position of the lipid molecule has a molecular formula of $C_{16}H_{32}O_2$, while the fatty acid occupying the other two positions has a molecular formula of $C_{18}H_{34}O_2$.
- 2 More ATP can be obtained from the lipid molecule than equal mass of glycogen.
- 3 The molecule is able to dissolve in an aqueous medium due to the presence of polar bonds.
- 4 The molecule increases fluidity of membranes as it contains a kink.

- A** 1 and 4
B 2 and 3
C 3 and 4
D 1, 3 and 4

- 4 The diagram shows the cell surface membrane of an actively respiring liver cell placed in a solution of glucose with a lower water potential than that of the liver cells. Arrows **P**, **Q**, **R** and **S** show the movement of various particles across the cell surface membrane.



Which arrows correctly reflect the movements of oxygen, carbon dioxide, glucose and water across the surface membrane of this cell?

	Oxygen	Carbon dioxide	Glucose	Water
A	P	Q	R	S
B	Q	S	R	S
C	Q	S	P	P
D	R	S	P	R

- 5 Human hair is made up from bundles of a fibrous protein called keratin, which is similar to collagen.

The diagram shows the folded structure of keratin polypeptide chain, which forms the basic unit of hair fibre structure.

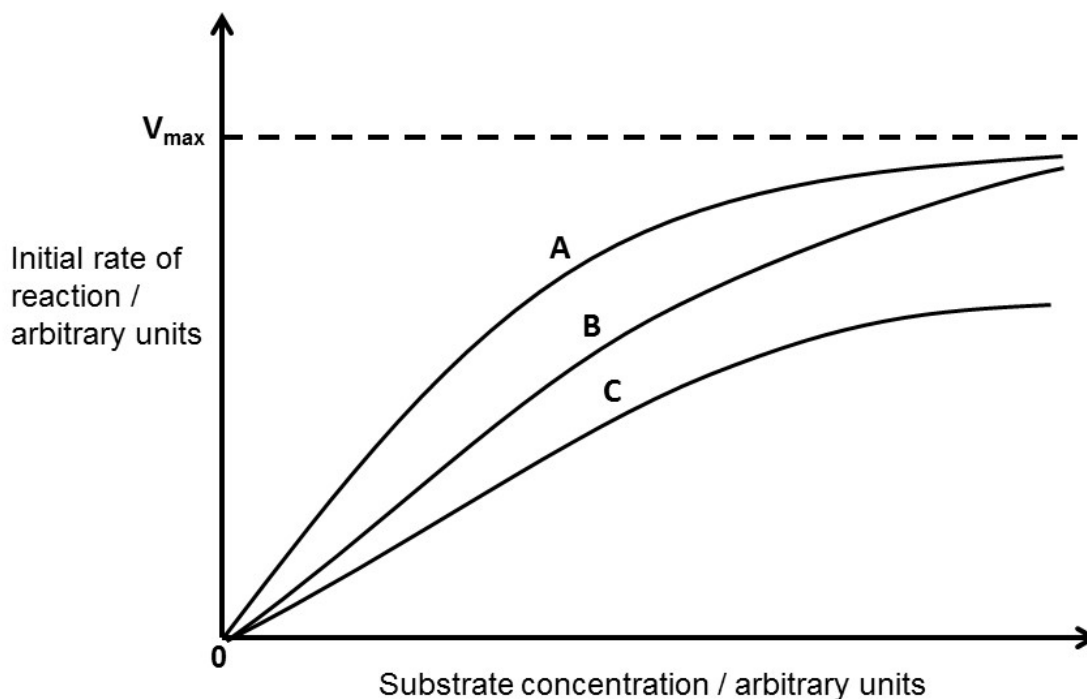


When hair is wet, it can lengthen temporarily due to the breaking of bonds in the folded keratin polypeptide chain. When it dries, the hair gradually returns to its original length.

What are broken to produce this length of the keratin chain structure?

- A** peptide bonds
- B** disulfide bonds
- C** hydrophobic interactions
- D** hydrogen bonds

- 6 The diagram shows the effect of increasing substrate concentration on the rate of an enzyme-catalysed reaction at different concentrations of an inhibitor of the enzyme.



A: no inhibitor added

B: 0.05% inhibitor added

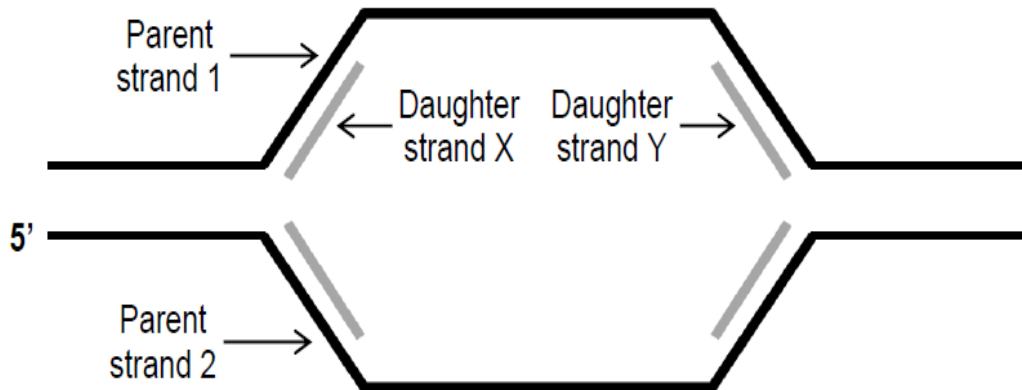
C: 0.1% inhibitor added

Which statements correctly explain the results of this investigation?

- 1 The inhibitor is a competitive inhibitor.
- 2 The inhibitor is a non-competitive inhibitor.
- 3 Adding 0.1% inhibitor lowers the effective enzyme concentration by altering enzyme tertiary structure, therefore the maximal rate of reaction cannot be restored at high substrate concentration.
- 4 Increasing inhibitor concentration results in the inhibitor being able to compete effectively with the substrate molecules to form more enzyme – inhibitor complexes, therefore higher concentration of inhibitor leads to rate of reaction not reaching maximum at high substrate concentration.

- A** 1 and 4
B 1 and 3
C 2 and 3
D 2 and 4

- 7 A simplified representation of a replication bubble is shown in the figure below. Parental strands 1 and 2 and the growing daughter strands X and Y are indicated.



Which of the following statements about the synthesis of daughter strands X and Y is correct?

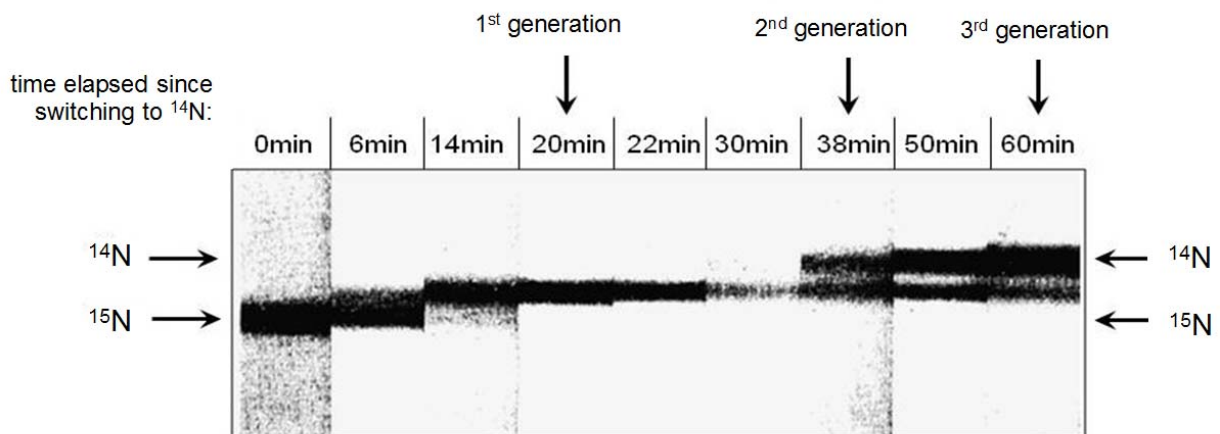
- A Daughter strands X and Y are synthesised away from their respective replication forks.
- B Daughter strand X is synthesised continuously while daughter strand Y is synthesised in the form of Okazaki fragments.
- C Daughter strand X is synthesised in the $5' \rightarrow 3'$ direction while daughter strand Y is synthesised in the $3' \rightarrow 5'$ direction.
- D To synthesise daughter strands X and Y, both parental strands 1 and 2 are read in the $3'$ to $5'$ direction.

- 8 In an investigation to study the mode of DNA replication, *Escherichia coli* (*E. coli*) cells were grown in a nutrient medium containing heavy isotope of nitrogen (^{15}N) for an extended period of time until all the DNA was labelled.

These *E. coli* cells were then transferred to a nutrient medium containing only light isotope of nitrogen (^{14}N) and were allowed to multiply over three generations. The DNA of the *E. coli* cells was then harvested at nine different time intervals.

Subsequently, density gradient centrifugation of these *E. coli* DNA using caesium chloride was performed.

The diagram shows the results obtained.



Which statements are consistent with the results observed?

- 1 The generation time for *E. coli* is about 20 minutes.
 - 2 In the 1st generation, only hybrid $^{14}\text{N}/^{15}\text{N}$ DNA was produced.
 - 3 In the 3rd generation, 75% hybrid $^{14}\text{N}/^{15}\text{N}$ DNA and 25% light $^{14}\text{N}/^{14}\text{N}$ DNA were produced.
 - 4 In the subsequent 4th generation, only light $^{14}\text{N}/^{14}\text{N}$ DNA would be produced.
- A 1 and 2
 B 2 and 3
 C 1 and 4
 D 1, 3 and 4

- 9 The coding strand of part of a DNA molecule has the sequence 5' GAATTA 3'.

Which row is correct for this part of the corresponding template, mRNA and tRNA?

	the sequence of the template (non-coding DNA sequence)	the sequence of the mRNA molecule	the anticodons of tRNA used in translation
A	5' TAATTC 3'	5' CUUAUU 3'	5' CUU 3' and 5' AAU 3'
B	5' TAATTC 3'	5' GAAUUA 3'	5' UAA 3' and 5' UUC 3'
C	5' CTTAAT 3'	5' CUUAUU 3'	5' CUU 3' and 5' AAU 3'
D	5' CTTAAT 3'	5' GAAUUA 3'	5' UAA 3' and 5' UUC 3'

- 10 Which of the statements are **false** about dengue virus and HIV?

- 1 The virus must contain a RNA polymerase.
- 2 The virus must encode protein for entry into host cells.
- 3 Replication of their genetic material must occur in the nucleus of the host cell.
- 4 The virus must encode proteases which can generate functional proteins from a polyprotein.

- A** 4 only
B 1 and 3
C 2 and 3
D 2, 3 and 4

- 11 Which correctly describes the transfer of DNA from one bacterium to another?

- 1 In transformation, bacterial cell takes up foreign DNA from culture medium.
- 2 In generalised transduction, specific bacterial genes are transferred by bacteriophage from one bacterium to another.
- 3 In conjugation, DNA replication of the F plasmid occurs in both donor and recipient cells.
- 4 Binary fission always allows chromosomal and non-chromosomal DNA to be transferred.

- A** 1 and 3
B 2 and 3
C 1, 3 and 4
D 1, 2 and 4

- 12** X-gal is a compound which turns from colourless to blue, when cleaved by the enzyme beta-galactosidase. This enzyme is coded for by the *lac Z* structural gene, which is part of the *lac* operon in *E.coli*.

Which of the following mutations would result in the formation of blue colonies when *E.coli* cells are cultured in a medium containing X-gal but deficient in lactose and glucose?

- 1 A gain-of-function mutation in the gene coding for the catabolite activator protein.
- 2 A loss-of-function mutation in promoter region of the *lac I* regulatory gene.
- 3 A loss-of-function mutation which affects the DNA binding site of the Lac repressor.
- 4 A mutation which leads to the loss of the cAMP binding sites of catabolite activator protein.

- A** 2 and 3
B 1 and 4
C 2, 3 and 4
D 1, 2, 3 and 4

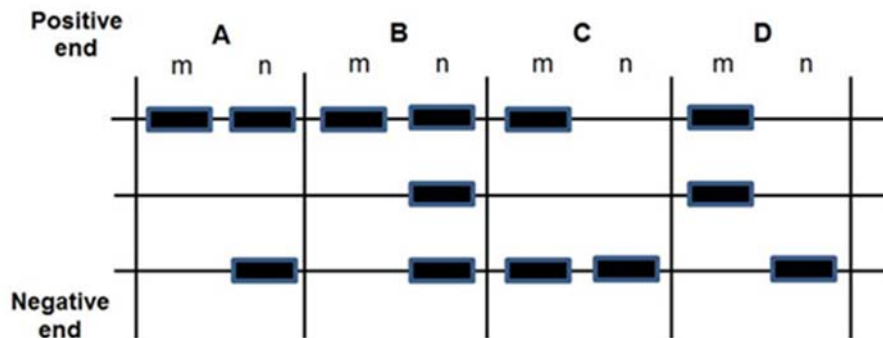
- 13** Which statement best explains how related genes involved in the same metabolic pathway are expressed together in eukaryotic cells?

- A** Related genes are usually located on the same chromosome so that they can be controlled by the same set of control elements.
- B** The same set of transcription factors may be capable of recognising the same promoter site of related genes.
- C** The same sets of control elements are associated with related genes so that they can be recognised by the same sets of transcription factors.
- D** Within the control element of related genes, the specific numbers of transcription factors binding to the control element will enable related genes to be expressed.

- 14** There are two forms of the seeds of the garden pea, *Pisum sativum*, one with a wrinkled skin and the other with a smooth skin. Plants with wrinkled seeds have a recessive mutation caused by the insertion of a transposable element into a gene.

Two heterozygous pea plants were crossed. Two of the offspring (**m** and **n**) had the DNA for this gene removed and separated by electrophoresis.

Which pattern of bands shows a wrinkled phenotype and a smooth phenotype?



- 15** Tay-Sachs disease is a fatal neurodegenerative disease which is caused by a mutation in the hexosaminidase A (Hex A) gene located on chromosome 15.

Part of the sequence of the non-template (coding) DNA strand of the normal Hex A allele and the mutated Tay-Sachs allele are shown below. The sequences are the same as the mRNA sequence of both alleles.

DNA sequences of normal Hex A allele:

Amino acid position		424	425	426	427	428	429	430	431	
Non-template DNA	5'...	CGT	ATA	TCC	TAT	GGC	CCT	GAC	TGT	...3'

DNA sequences of mutated Tay-Sachs allele:

Amino acid position		424	425	426	427	428	429	430	431	
Non-template DNA	5'...	CGT	ATA	TCT	ATC	CTA	TGG	CCC	TGA	...3'

For both alleles, 9 different amino acids are encoded for by the DNA triplets:

Amino acid	DNA triplet	Amino acid	DNA triplet
Arg	CGT	Leu	CTA
Asp	GAC	Pro	CCC, CCT
Cys	TGG, TGT	Ser	TCC, TCT
Gly	GGC	Tyr	TAT
Ile	ATA, ATC	Stop codon	TAG, TAA, TGA

Which statement is true?

- A** The disease is caused by the deletion of one DNA nucleotide.
- B** The Hex A protein encoded for by the Tay-Sachs allele is non-functional due to a frameshift mutation.
- C** The polypeptide encoded for by the Tay-Sachs allele has the same number of amino acids as that encoded by the normal Hex A allele.
- D** At amino acid position 431, there is a silent mutation.

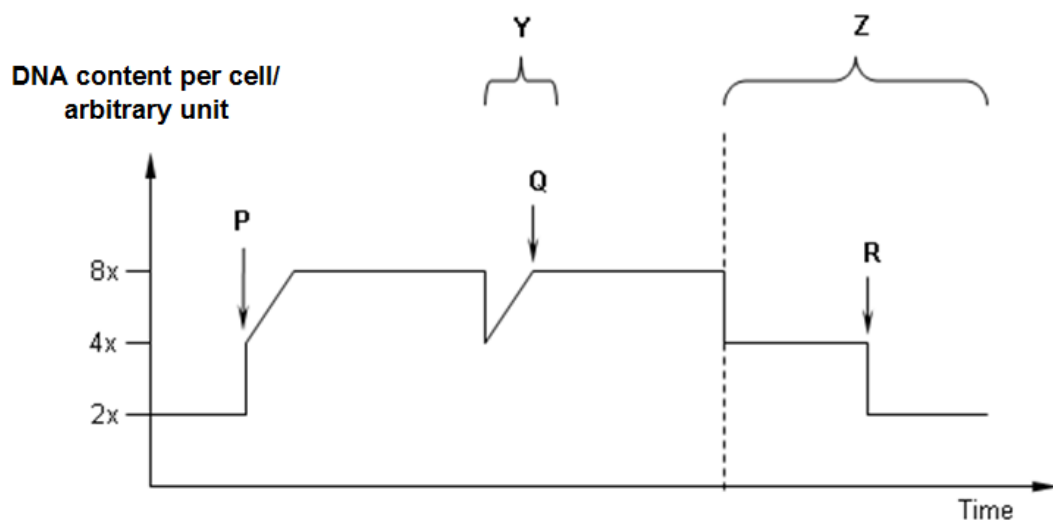
16 Edwards' syndrome is a common autosomal disorder caused by chromosome aberration.

- Edwards' syndrome is a trisomy of chromosome 18 that affects all cells.
- The extra chromosome is most often of maternal origin.
- In approximately more than 50% of individuals diagnosed with the syndrome, two of the three chromosomes 18 present are found to be nearly genetically identical.

What can be correctly concluded from this information?

- A** Non-disjunction of chromosome 18 commonly occurs during gamete formation in females.
- B** Non-disjunction of chromosome 18 occurs more frequently during meiosis I than in meiosis II.
- C** Non-disjunction of chromosome 18 occurs most commonly in female embryos.
- D** Two of the three chromosomes 18 are nearly genetically identical due to uneven crossing over resulting in chromosomal translocation.

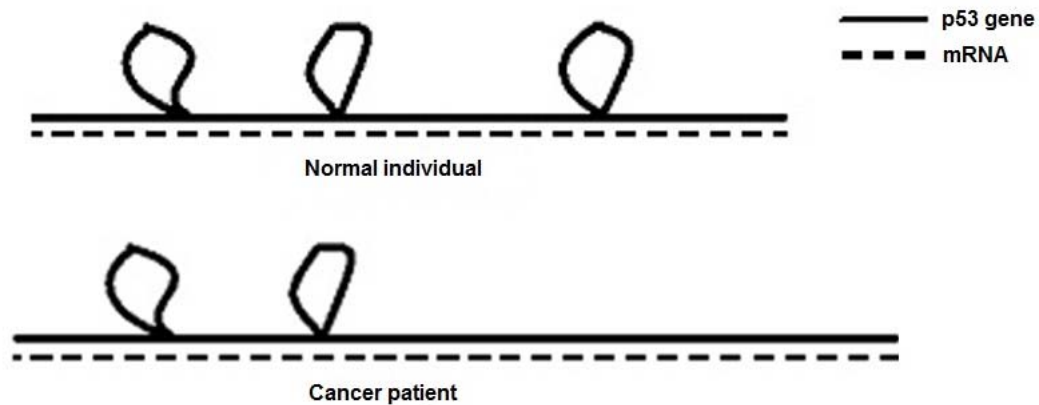
17 The graph represents the changes in the DNA content within a cell at different stages in the cycle.



Name the events occurring at **P**, **Q**, **R** and identify the stage where meiosis is occurring.

	P	Q	R	Meiosis occurring at
A	S phase	fertilisation	cytokinesis	Y
B	fertilisation	interphase	cytokinesis	Z
C	S phase	prophase	telophase	Y
D	fertilisation	metaphase	Telophase	Z

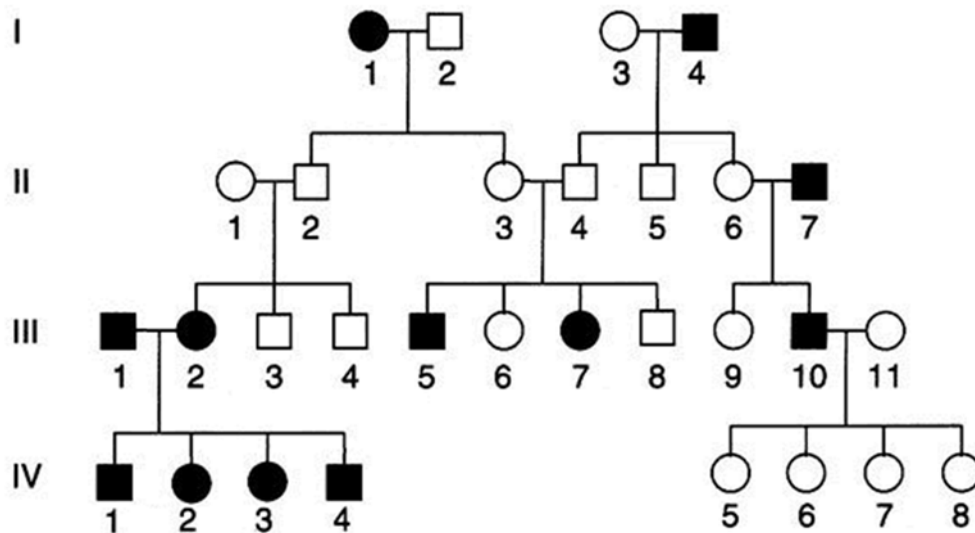
- 18 mRNA was isolated from a normal individual and a patient suffering from cancer. The mRNA was allowed to hybridise with the *p53* DNA gene sequence. The schematic diagram shows the results of the hybridisation process under the electron microscope.



Which of the following could be a possible explanation why the patient is suffering from cancer?

- A A point mutation had occurred in the intron leading to the failure to excise one intron, resulting in a longer dysfunctional protein being produced.
- B Gene amplification had occurred leading to the multiple copies of a trinucleotide repeat in an intron, hence causing splice site to be misread due to frameshift mutation, resulting in a longer dysfunctional protein being produced.
- C A point mutation had occurred in the intron leading to the failure to excise one exon, resulting in a hyperactive protein being produced.
- D A point mutation had occurred leading to the failure of spliceosome to recognise splice sites, causing the excision of the wrong intron, resulting in a degradation-resistant protein being produced.

19 In the pedigree chart below, the characteristic, adherent earlobes, were studied.



Which row shows the correct genotype(s) of III-3 and III-6, and the inheritance pattern?

	Genotype(s) of III-3 and III-6	Inheritance pattern
A	X^AY , X^AX^a or X^AX^A	Sex-linked recessive
B	Aa	Autosomal dominant
C	X^aY , X^aX^a	Sex-linked dominant
D	AA or Aa	Autosomal recessive

20 The table shows the results of a series of crosses in a species of small mammal.

coat colour phenotype		
male parent	female parent	offspring
dark grey	light grey	dark grey, light grey, albino
light grey	albino	light grey, white with black patches
dark grey	white with black patches	dark grey, light grey
light grey	dark grey	dark grey, light grey, white with black patches

What explains the inheritance of the range of phenotypes shown by these crosses?

- A** One gene with a pair of co-dominant alleles
- B** One gene with multiple alleles
- C** Sex-linkage of the allele for grey coat colour
- D** Two genes, each with a dominant and recessive allele

- 21** Fruit colour in a particular type of plant is controlled by two genes, C/c and G/g. The phenotypes of progeny that are expected from crossing of two plants, each of which is heterozygous at both gene loci, are shown in the Punnett square.

Gametes	CG	Cg	cG	cg
CG	white	white	white	white
Cg	white	white	white	white
cG	white	white	yellow	yellow
cg	white	white	yellow	green

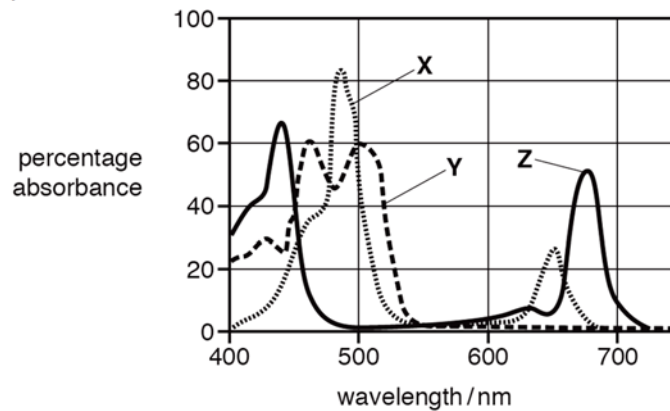
What best explains the proportion of white to yellow to green fruit colour in the Punnett square?

- A** dominant epistasis in which the epistatic allele is C
- B** dominant epistasis in which the epistatic allele is G
- C** recessive epistasis in which the epistatic allele is c
- D** recessive epistasis in which the epistatic allele is g

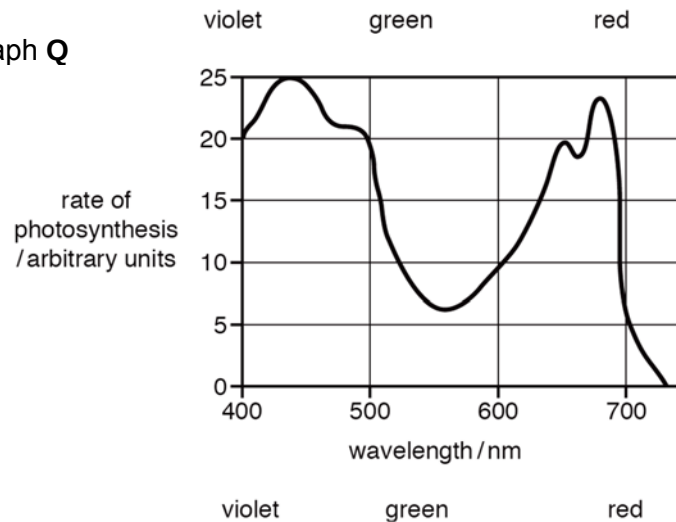
- 22 Graph **P** shows the absorption spectra of three types of photosynthetic pigment, **X**, **Y** and **Z**, extracted from the leaves of a flowering plant. **X** is chlorophyll b.

Graph **Q** shows the action spectrum for photosynthesis for the same plant.

Graph **P**



Graph **Q**



Five students were asked to relate the information shown in graphs **P** and **Q** to their knowledge and understanding of the light-dependent stage of photosynthesis.

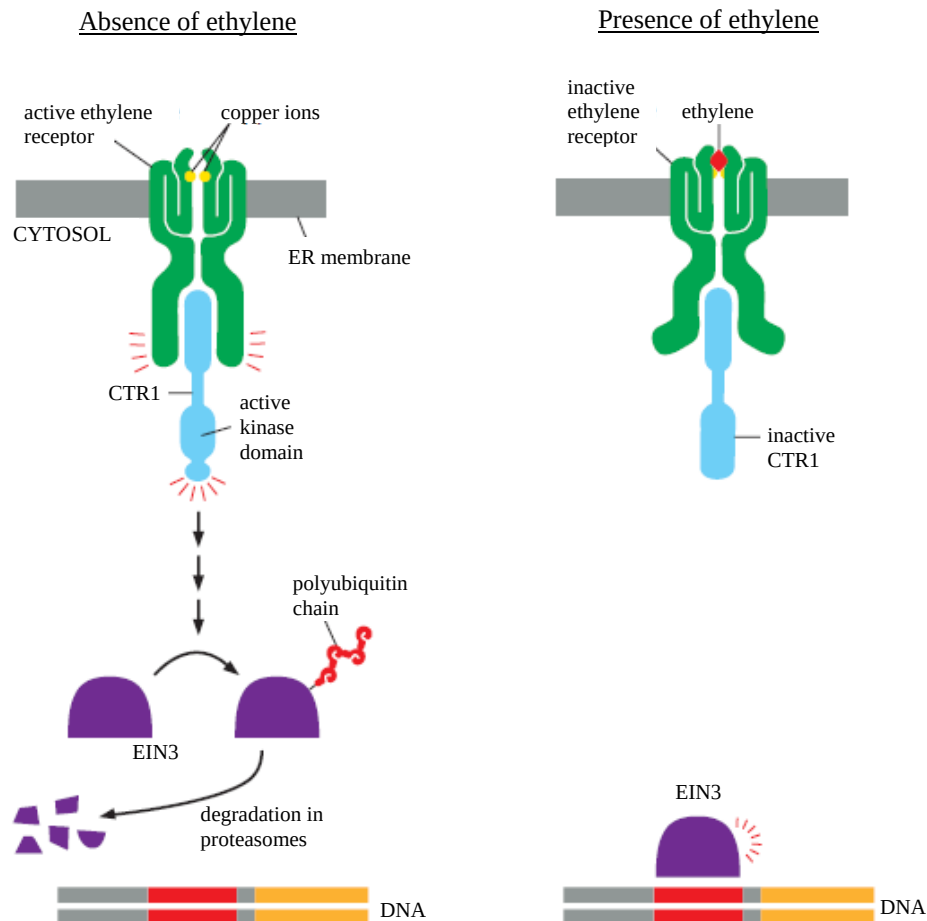
student	comment
1	The low rate of photosynthesis in green light suggests that more green light is reflected than absorbed by the three pigments.
2	The poor absorption of green light by all three pigment types will provide only enough energy for cyclic photophosphorylation to occur.
3	The presence of pigment Y extends the ability of the plant to absorb light in the blue-green part of the spectrum but not the yellow-green part of the spectrum.
4	Non-cyclic photophosphorylation occurs at a wavelength of 700 nm, indicating that pigment Y is more likely to be chlorophyll a than pigment Z .

Which students made biologically correct comments?

- A** 1 and 3 **B** 1 and 4
C 2 and 3 **D** 2 and 4

- 23** Ethylene gas is a plant hormone that regulates plant growth, development and response to environmental stress. It is produced from leaves, roots, stems, flowers and especially ripened fruits.

Plants have various ethylene receptors which are located in the endoplasmic reticulum (ER). The diagram shows the ethylene signalling pathway. Ethylene receptors are dimeric, transmembrane proteins, with a copper-containing ethylene-binding domain and a domain that interacts with a cytoplasmic protein called CTR1.



Which statements provide the most direct evidence that the ethylene gas signalling mechanism functions to mediate gene expression?

- 1 In the absence of ethylene, active CTR1 stimulates the ubiquitination and degradation in proteasomes of EIN3.
- 2 In the absence of ethylene, the active ethylene receptors halt transcription of ethylene-responsive genes through degradation of EIN3.
- 3 In the presence of ethylene, its binding inactivates the receptor, altering their conformation so that they no longer activate CTR1.
- 4 In the presence of ethylene, the EIN3 protein does not undergo selective degradation and can now activate the transcription of the large number of ethylene-responsive genes.

A 1 and 2

B 2 and 3

C 2 and 4

D 3 and 4

24 What is the main purpose of the second messengers in signal transduction pathways?

- A** They allow for long distance signalling between cells by being activated when they bind to extracellular molecules.
- B** They amplify the signal without involving protein kinase cascade.
- C** They can always cross the membranes of the organelles to initiate cellular processes.
- D** They relay a signal from the plasma membrane to the cytoplasm by sudden increase of their cytosolic concentration.

25 Which of the following shows the correct sequence of events?

1	adaptation of a population	→	competition and predation leading to natural selection	→	behavioural isolation	→	allopatric speciation
2	adaptation of a population	→	competition and predation leading to natural selection	→	physiological isolation	→	allopatric speciation
3	competition and predation leading to natural selection	→	physiological isolation	→	adaptation of isolated populations	→	sympatric speciation
4	competition and predation leading to natural selection	→	geographical isolation	→	adaptation of isolated populations	→	allopatric speciation

- A** 1 and 2
- B** 2 and 3
- C** 3 and 4
- D** 1, 3 and 4

- 26** Myxomatosis is a viral disease of rabbits. It spreads rapidly and most rabbits die within 14 days of being infected. Myxomatosis has been deliberately used to reduce the number of rabbits in countries where they are a significant crop pest.

The initial release of the virus caused populations to fall by over 90%. Resistance to myxomatosis increased in the 70 years following initial release, so at present time, up to 50% of infected rabbits are able to survive.

Which statement could explain the increasing frequency of resistance to myxomatosis in the years following release of the virus?

- 1 In populations with high incidences of myxomatosis, mutations leading to resistance are more likely to occur.
- 2 Infected rabbits die quickly, hence the alleles that code for myxomatosis are eliminated from the population.
- 3 The initial release of the virus led to a bottleneck event, greatly altering the frequency of alleles in rabbit populations.
- 4 During disease outbreaks there is greater food availability for the surviving rabbits, increasing the probability that they survive and breed.

- A** 4 only
B 1 and 2
C 2 and 4
D 2, 3 and 4

- 27** Cabbage, *Brassica oleracea* ($2n = 20$), and radish, *Raphanus sativus* ($2n = 18$), are different species of the Brassicaceae family. When these plants are crossed, a hybrid is produced. Two cells from the hybrid plant are fused to form a single cell which is grown into a plant.

Which of the following is true?

- A** The hybrid plant is a polyploidy.
B The hybrid plant can produce gametes.
C The single fusion cell may eventually result in a new plant species.
D The plant that arises from the single fusion cell may be crossed with either *Brassica oleracea* or *Raphanus sativus* to produce a viable and fertile offspring.

28 Rifampicin is a major antibiotic used in treating *Mycobacterium tuberculosis* infections.

- Rifampicin binds to β subunit of prokaryotic RNA polymerase, forming a stable drug – protein complex.
- The β subunit of prokaryotic RNA polymerase forms a part of the site involved in RNA synthesis and interaction with DNA.
- Prokaryotic RNA polymerase is a complex made up of four major subunits coded for by four genes, rpoA, rpoB, rpoC and rpoD.
- Mutations in the rpoB gene, which codes for the β subunit, results in decreased affinity of the β subunit for Rifampicin.

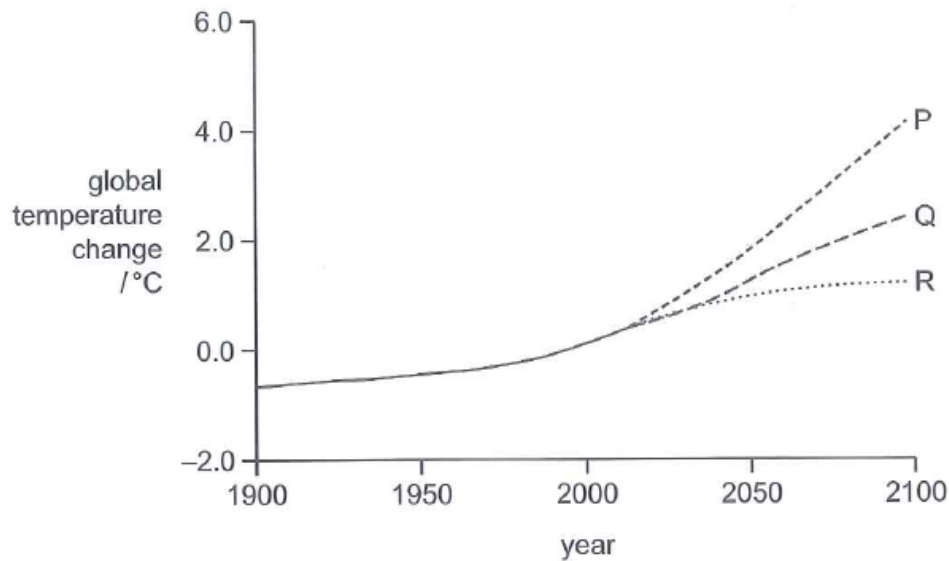
It was observed that some Rifampicin – resistant bacteria have a lower rate of reproduction than bacteria susceptible to Rifampicin while other Rifampicin – resistant bacteria have the same rate of reproduction as bacteria susceptible to Rifampicin.

What could account for the above observation?

- 1 Mutation of rpoB gene gives bacteria resistance to Rifampicin but also affects the rate of transcription.
 - 2 Rifampicin – resistant bacteria which reproduction rate is unaffected are heterozygous for the rpoB gene, therefore producing sufficient copies of fully functional RNA polymerases.
 - 3 Gain of function mutations occur in rpoA, rpoC and rpoD genes, compensating the effect of rpoB gene mutation on the function of RNA polymerase.
 - 4 Mutations can occur at different points in the rpoB gene sequence, resulting in a range of effects on the function of RNA polymerase and confer different extents of Rifampicin resistance to bacteria.
- A** 1 and 2
B 2 and 3
C 1, 3 and 4
D 1, 2, 3 and 4

- 29** Which of the following statements correctly describes the relationship between macrophages and CD4 T cells?
- A** Macrophages secrete cytokines to cause CD4 T cells to divide and differentiate into T helper cells. These effector T cells in turn secrete cytokines to stimulate an increase in toxic chemical production to kill ingested bacteria.
 - B** Macrophages secrete chemokines to recruit circulating T helper cells to the infection site and these effector T cells secrete granzymes and perforin to induce apoptosis in macrophages which ingested bacteria, causing destruction of pathogens.
 - C** Macrophages act as antigen – presenting cells to activate naïve CD4 T cells to divide and differentiate into T helper cells. These effector T cells in turn secrete chemokines to recruit more macrophages to accumulate at the site of infection to destroy pathogens via phagocytosis and toxic chemical production.
 - D** Macrophages act as antigen – presenting cells to activate naïve CD4 T cells to divide and differentiate into cytotoxic T cells. These effector T cells in turn secrete granzymes and perforin to induce apoptosis in macrophages which ingested bacteria, causing destruction of pathogens.

- 30 The graph shows the predicted change in global temperature using three different models, **P**, **Q** and **R**. Model **Q** assumes that no new factors act to influence the rate of climate change.



The predictions based on models **P** and **R** can be explained using some of the following statements.

- 1 An increased global temperature and reduced rainfall will lead to an increase in forest fires.
- 2 Permanently frozen soil and sediment in the Arctic will begin to thaw as global temperatures increase.
- 3 Rising sea temperatures will cause increased growth of photosynthetic algae.
- 4 Rising sea temperatures will reduce the solubility of greenhouse gases in the oceans.

Which of these statements support predictions **P** and **R**?

	statements that support prediction P	statements that support prediction R
A	1, 2 and 4	3
B	1 and 3	2 and 4
C	2	1, 3 and 4
D	3 and 4	1 and 2

Paper 1 Answer Scheme

Qns	Ans	Qns	Ans
1	C	16	A
2	C	17	B
3	D	18	A
4	B	19	D
5	D	20	B
6	A	21	A
7	D	22	A
8	A	23	C
9	B	24	D
10	B	25	C
11	C	26	A
12	A	27	C
13	C	28	C
14	C	29	C
15	B	30	A